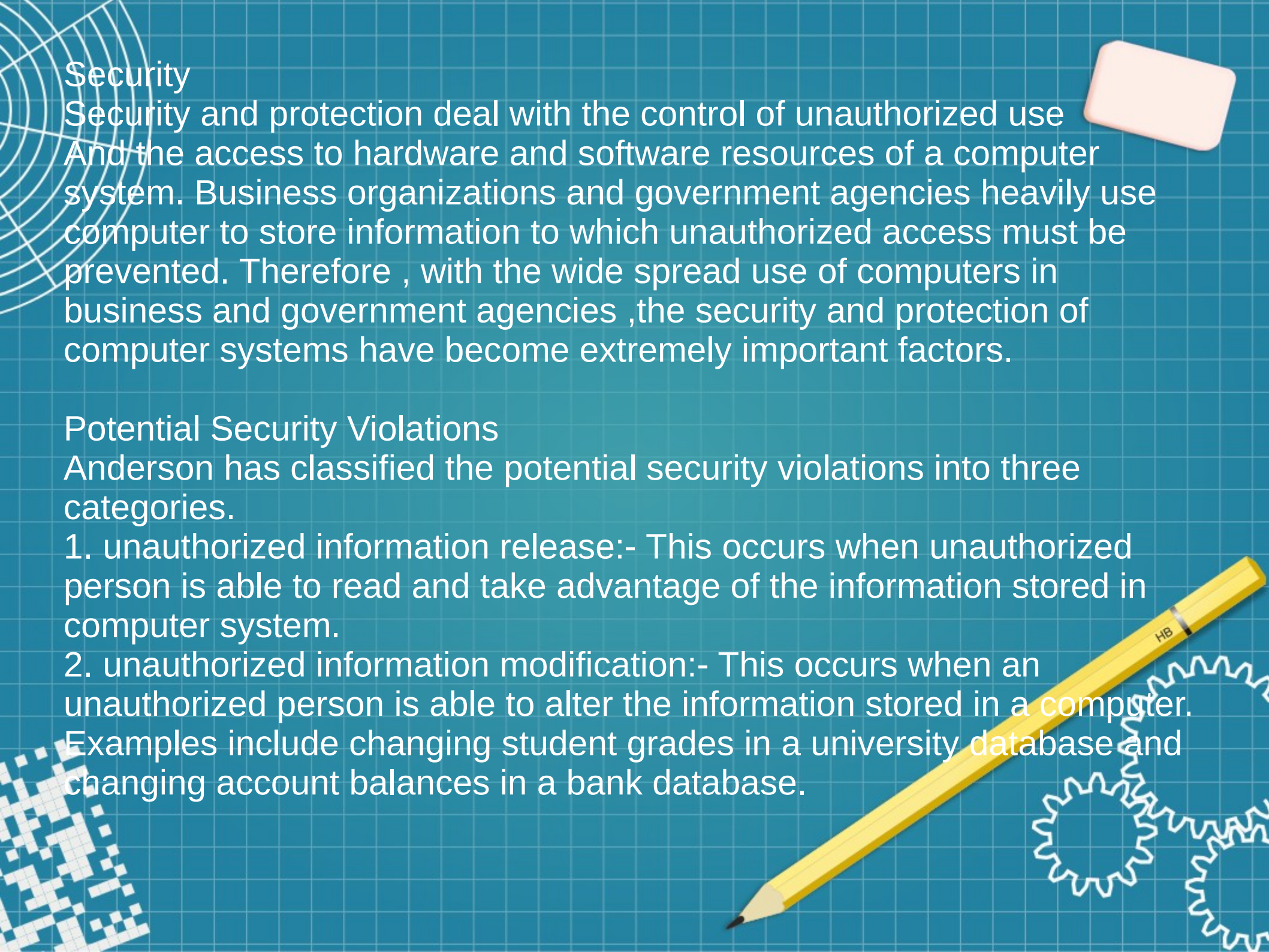




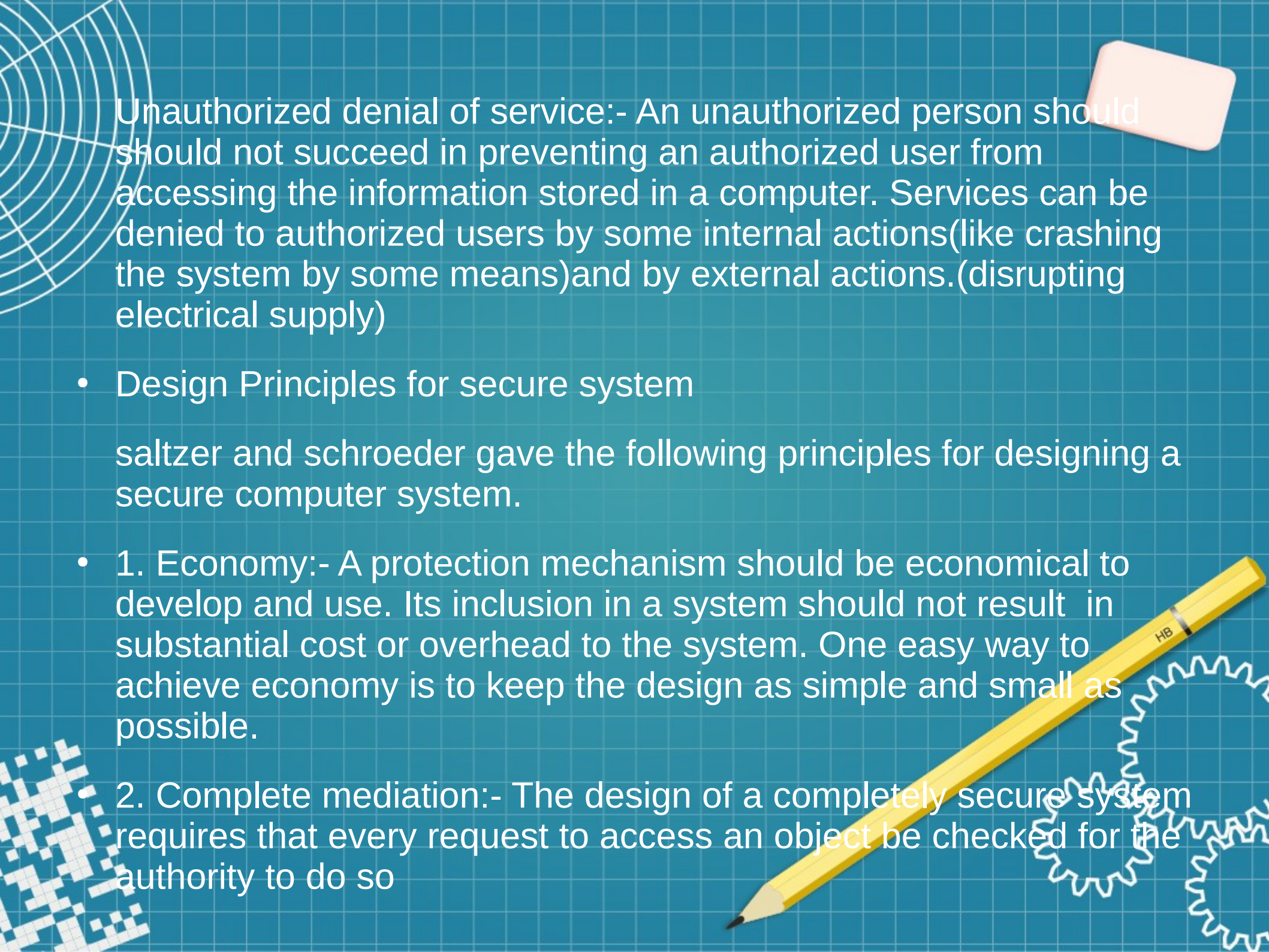
Security

Security and protection deal with the control of unauthorized use and the access to hardware and software resources of a computer system. Business organizations and government agencies heavily use computer to store information to which unauthorized access must be prevented. Therefore, with the wide spread use of computers in business and government agencies, the security and protection of computer systems have become extremely important factors.

Potential Security Violations

Anderson has classified the potential security violations into three categories.

1. unauthorized information release:- This occurs when unauthorized person is able to read and take advantage of the information stored in computer system.
2. unauthorized information modification:- This occurs when an unauthorized person is able to alter the information stored in a computer. Examples include changing student grades in a university database and changing account balances in a bank database.

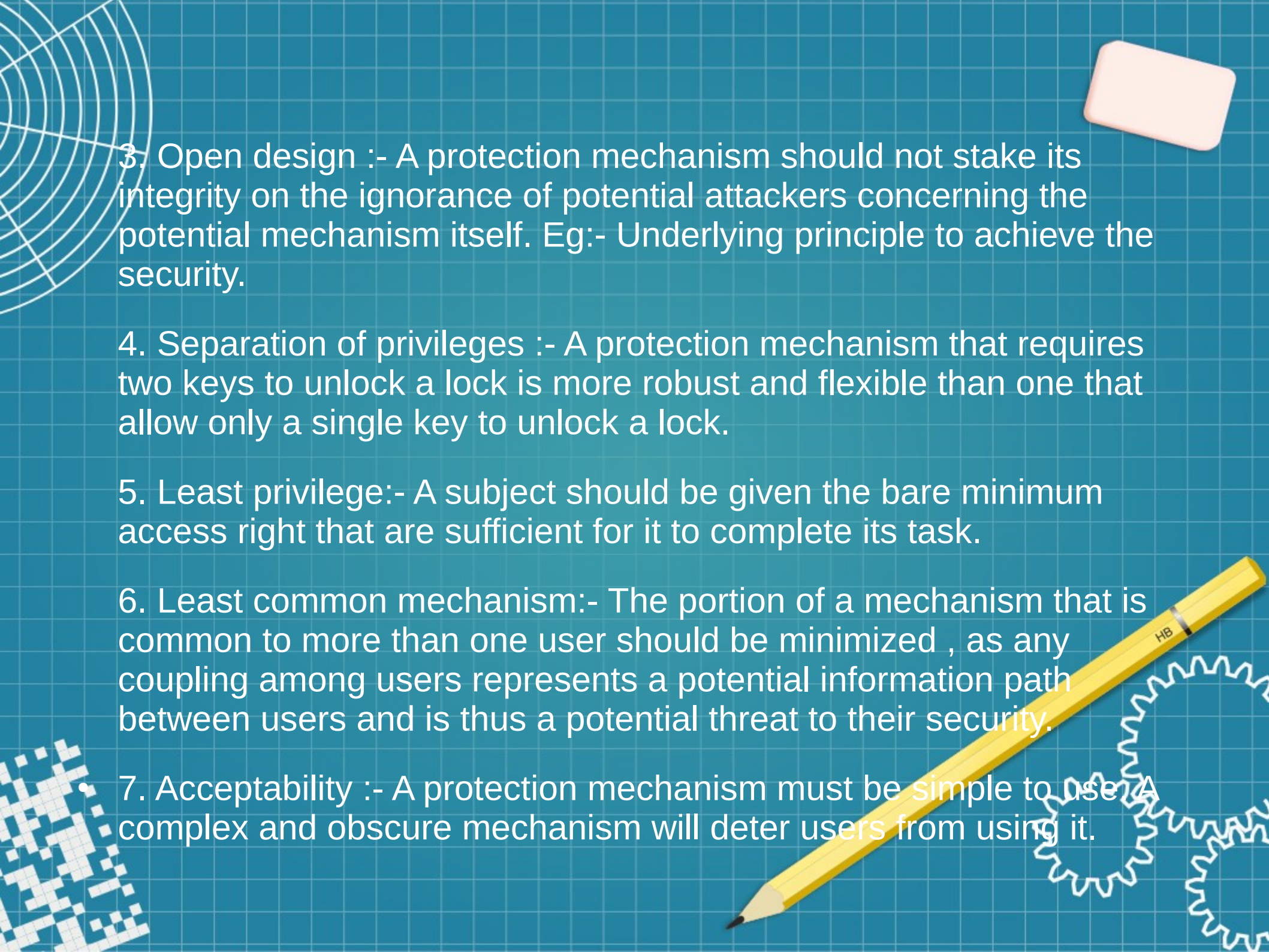


Unauthorized denial of service:- An unauthorized person should not succeed in preventing an authorized user from accessing the information stored in a computer. Services can be denied to authorized users by some internal actions (like crashing the system by some means) and by external actions. (disrupting electrical supply)

- Design Principles for secure system

saltzer and schroeder gave the following principles for designing a secure computer system.

- 1. Economy:- A protection mechanism should be economical to develop and use. Its inclusion in a system should not result in substantial cost or overhead to the system. One easy way to achieve economy is to keep the design as simple and small as possible.
- 2. Complete mediation:- The design of a completely secure system requires that every request to access an object be checked for the authority to do so



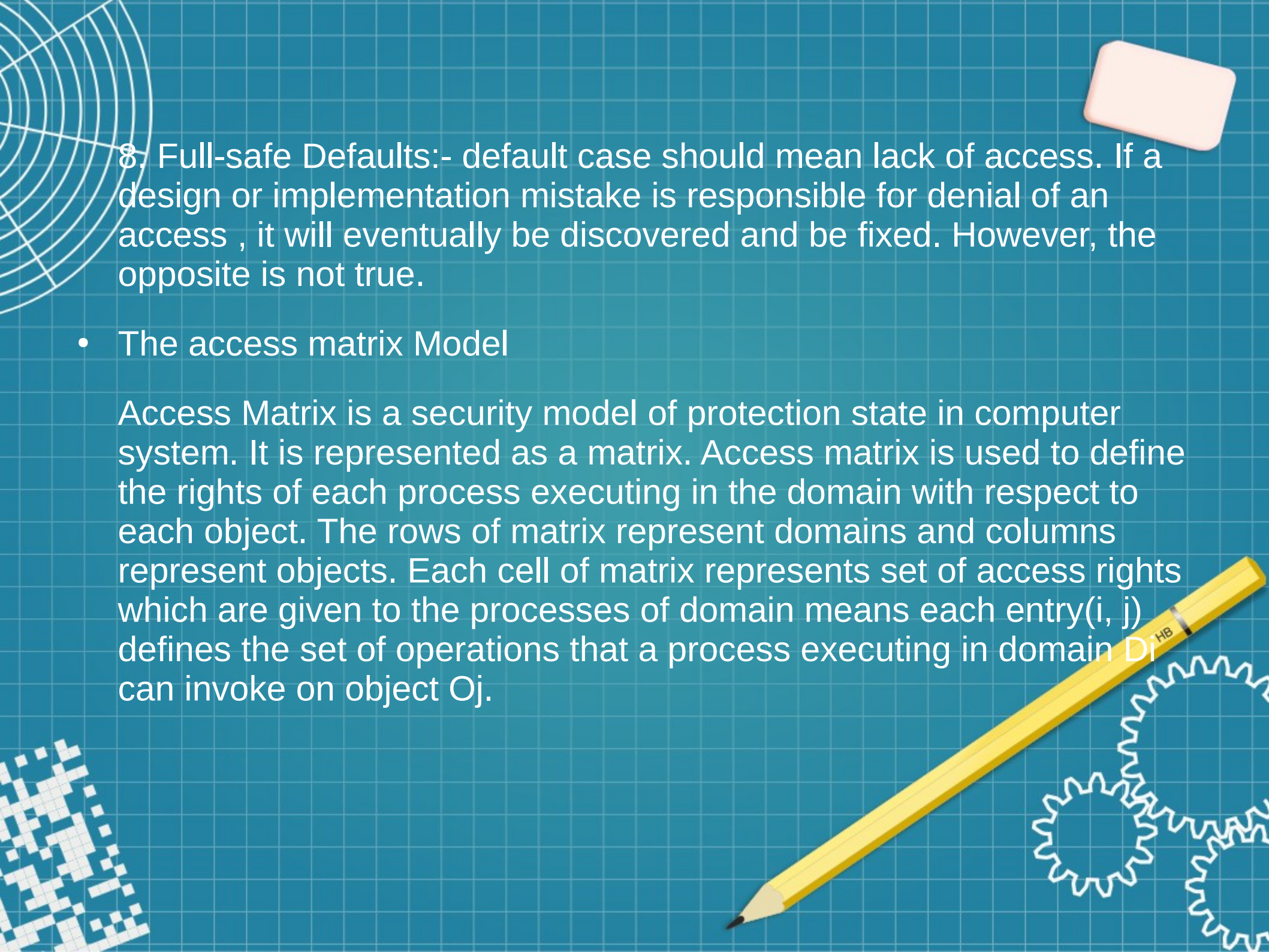
3. Open design :- A protection mechanism should not stake its integrity on the ignorance of potential attackers concerning the potential mechanism itself. Eg:- Underlying principle to achieve the security.

4. Separation of privileges :- A protection mechanism that requires two keys to unlock a lock is more robust and flexible than one that allow only a single key to unlock a lock.

5. Least privilege:- A subject should be given the bare minimum access right that are sufficient for it to complete its task.

6. Least common mechanism:- The portion of a mechanism that is common to more than one user should be minimized , as any coupling among users represents a potential information path between users and is thus a potential threat to their security.

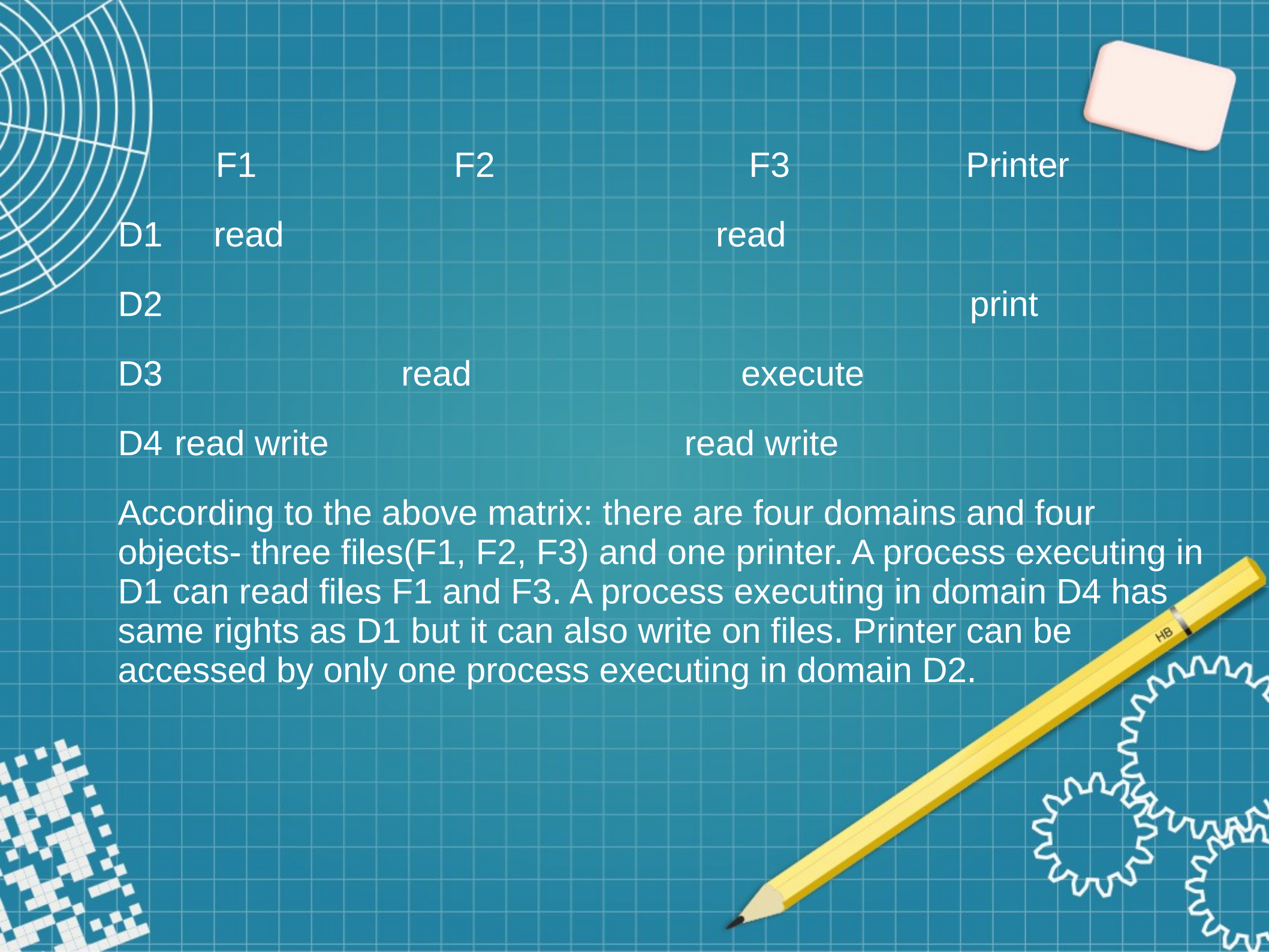
• 7. Acceptability :- A protection mechanism must be simple to use. A complex and obscure mechanism will deter users from using it.



8. Full-safe Defaults:- default case should mean lack of access. If a design or implementation mistake is responsible for denial of an access , it will eventually be discovered and be fixed. However, the opposite is not true.

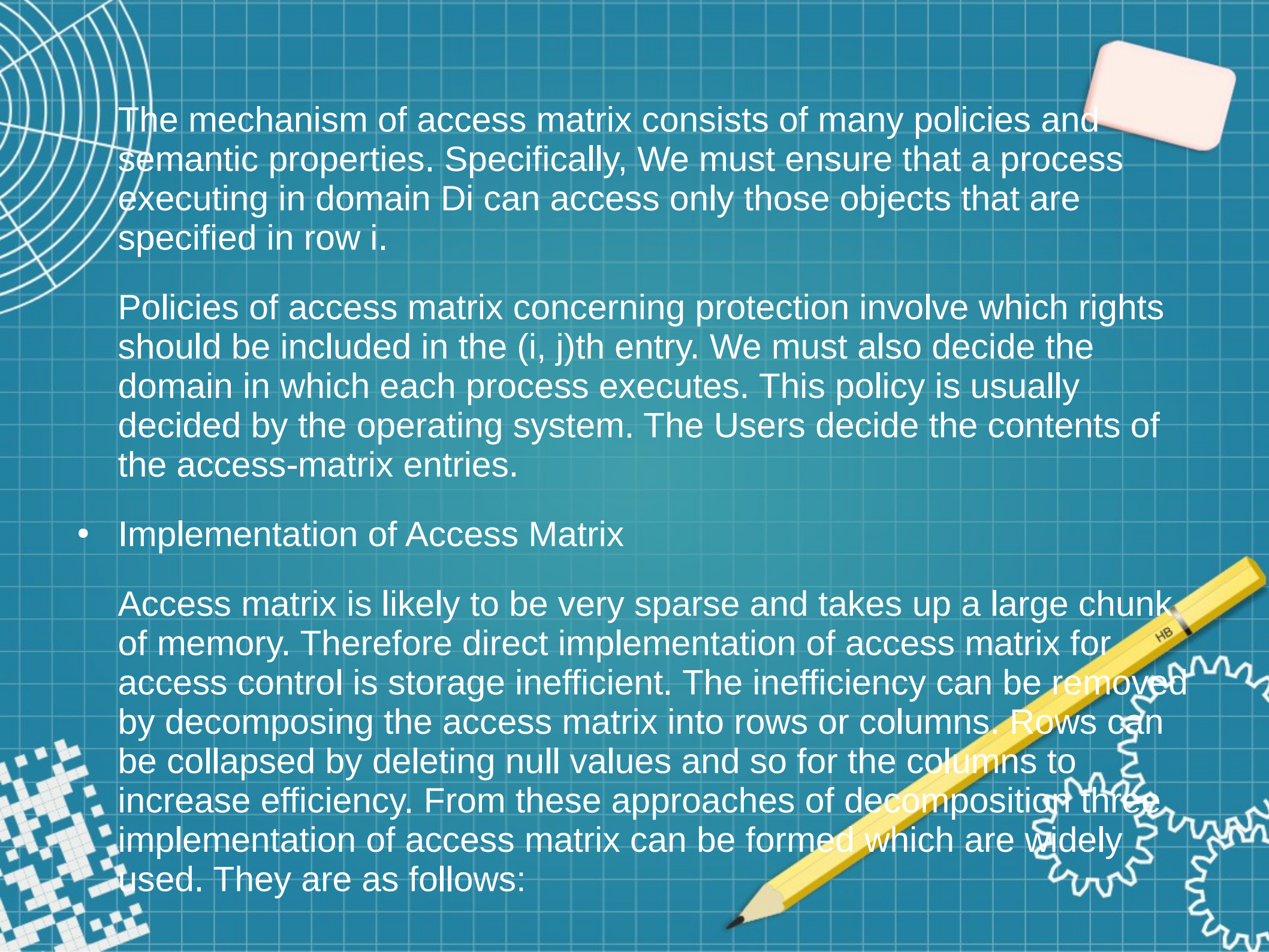
- The access matrix Model

Access Matrix is a security model of protection state in computer system. It is represented as a matrix. Access matrix is used to define the rights of each process executing in the domain with respect to each object. The rows of matrix represent domains and columns represent objects. Each cell of matrix represents set of access rights which are given to the processes of domain means each entry(i, j) defines the set of operations that a process executing in domain D_i can invoke on object O_j .



	F1	F2	F3	Printer
D1	read		read	
D2				print
D3		read	execute	
D4	read write		read write	

According to the above matrix: there are four domains and four objects- three files(F1, F2, F3) and one printer. A process executing in D1 can read files F1 and F3. A process executing in domain D4 has same rights as D1 but it can also write on files. Printer can be accessed by only one process executing in domain D2.

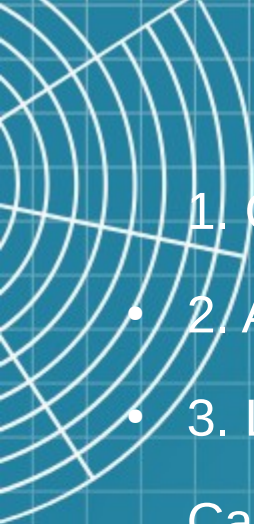


The mechanism of access matrix consists of many policies and semantic properties. Specifically, We must ensure that a process executing in domain D_i can access only those objects that are specified in row i .

Policies of access matrix concerning protection involve which rights should be included in the (i, j) th entry. We must also decide the domain in which each process executes. This policy is usually decided by the operating system. The Users decide the contents of the access-matrix entries.

- Implementation of Access Matrix

Access matrix is likely to be very sparse and takes up a large chunk of memory. Therefore direct implementation of access matrix for access control is storage inefficient. The inefficiency can be removed by decomposing the access matrix into rows or columns. Rows can be collapsed by deleting null values and so for the columns to increase efficiency. From these approaches of decomposition three implementation of access matrix can be formed which are widely used. They are as follows:



1. Capabilities

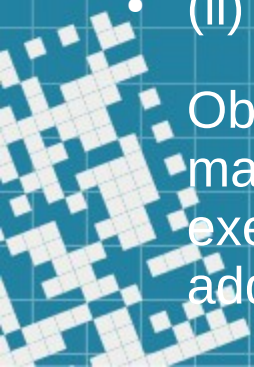
- 2. Access Control List
- 3. Lock and Key Method

Capabilities:

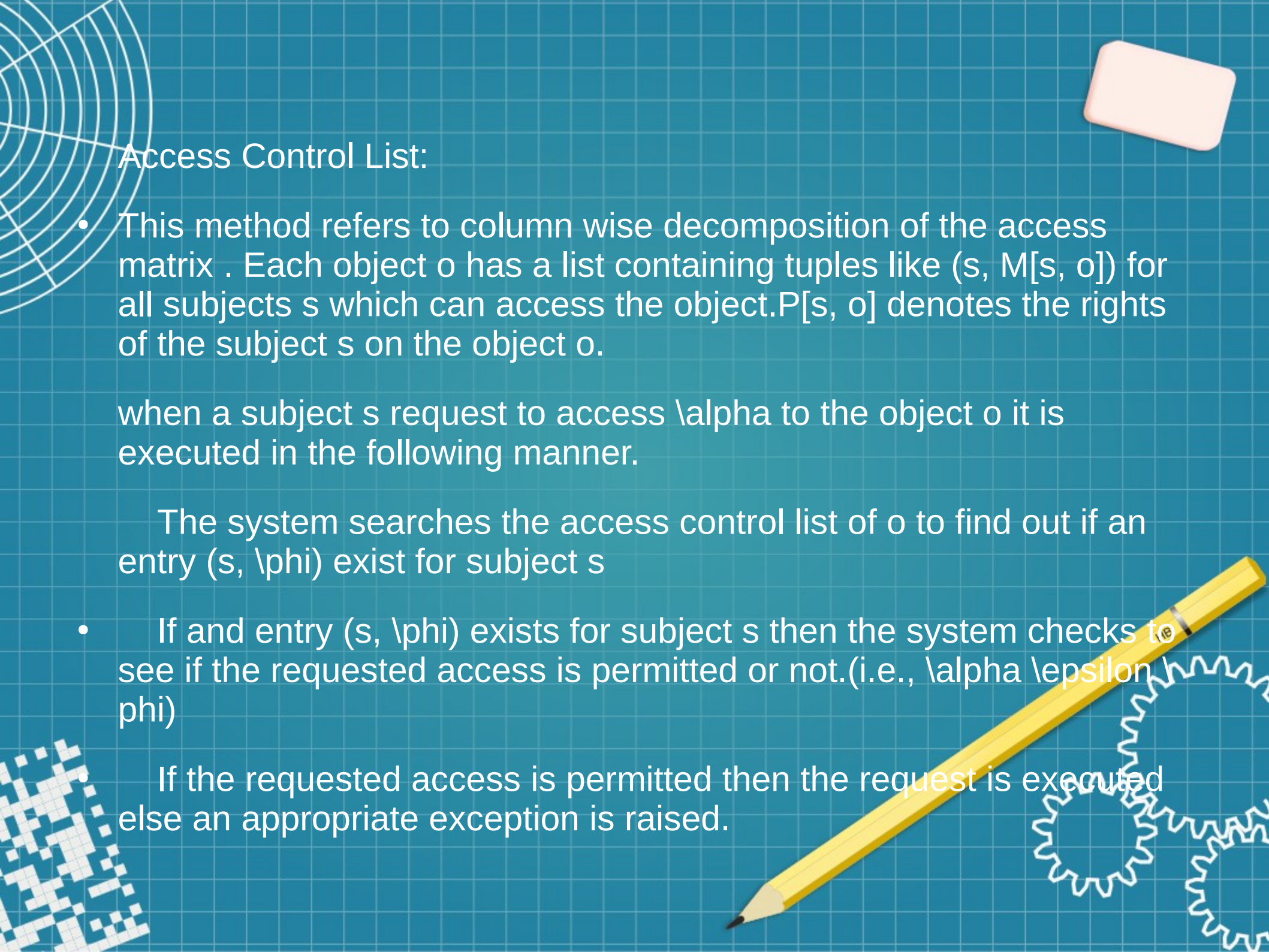
- This method refers to row wise decomposition of the access matrix. Each Subject is assigned with a list of tuples $(o, M[s, o])$ for all objects o that it is allowed to access. These tuples are called Capabilities. If a subject possesses a capability $(o, M[s, o])$ then it is allowed to access object o in the manner which is described in $M[s, o]$. A subject is allowed to access any objects for which it holds the capabilities. Capabilities are not meant to be forged.

Capabilities contain two fields:

- (i) Object Descriptor,
- (ii) Access Rights



Object Descriptor may contain the address of the objects and Access Rights may contain the rights which the subject has on object, mainly read, write, execute. Since object Descriptor contains address it may be used as an addressing mechanism also.



Access Control List:

- This method refers to column wise decomposition of the access matrix. Each object o has a list containing tuples like $(s, M[s, o])$ for all subjects s which can access the object. $P[s, o]$ denotes the rights of the subject s on the object o .

when a subject s request to access α to the object o it is executed in the following manner.

The system searches the access control list of o to find out if an entry (s, ϕ) exist for subject s

- If an entry (s, ϕ) exists for subject s then the system checks to see if the requested access is permitted or not.(i.e., $\alpha \in \phi$)
- If the requested access is permitted then the request is executed else an appropriate exception is raised.



Subjects Access Right

- ravi Read, Write, Execute
- rana Read
- jeffy Write
- alice Execute;

Easy revocation and Easy review of an access are the major feature of access control list.

- Lock and key Method:

The lock and key method is an hybrid of the access control list and capabilities method. In the lock and key method, every subject has a capability list that contains tuples of the form (o, key) , indicating the subject can access object o using key key . Objects has an access control list that contains tuples of the form $(lock, \phi)$, called a lock entry indicating lock $lock$ can be accessed by modes in the set

